

Name : Shan

Sap : 72387

BSAI

OOP (Lab)

◆ **LAB Task 12**

✧ **Q1: Abstortion Code Simple.**

➤ **Input**

✧ **Students.java**

```
package com.mycompany.graduatestudents;

public abstract class Students {

    abstract void grade();

    void show() {
        System.out.println("Shan is student");
    }
}
```

✧ **GraduateStudents.java**

```
package com.mycompany.graduatestudents;

public class GraduateStudents extends Students {

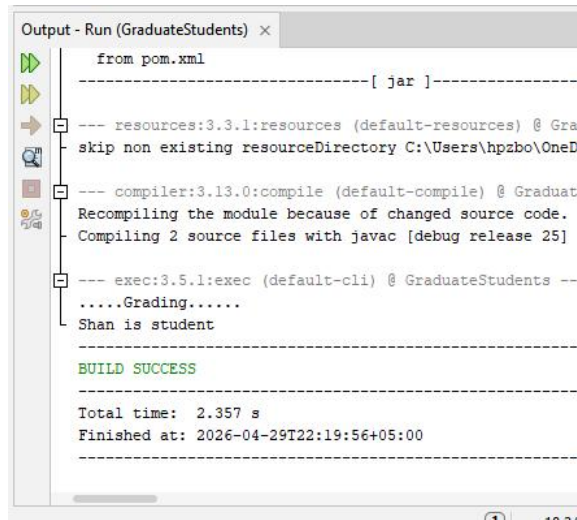
    void grade() {
        System.out.println(".....Grading.....");
    }

    public static void main(String[] args) {

        Students s = new GraduateStudents();

        s.grade();
        s.show();
    }
}
```

➤ Output



```
Output - Run (GraduateStudents) x
from pom.xml
-----[ jar ]-----
--- resources:3.3.1:resources (default-resources) @ Gra
skip non existing resourceDirectory C:\Users\hpzbo\OneD
--- compiler:3.13.0:compile (default-compile) @ Graduat
Recompiling the module because of changed source code.
Compiling 2 source files with javac [debug release 25]
--- exec:3.5.1:exec (default-cli) @ GraduateStudents --
.....Grading.....
Shan is student
-----
BUILD SUCCESS
-----
Total time: 2.357 s
Finished at: 2026-04-29T22:19:56+05:00
-----
```

- ✧ **Q2: Write a Java program to create an abstract class Employee with abstract methods calculateSalary() and displayInfo(). Create subclasses Manager and Programmer that extend the Employee class and implement the respective methods to calculate salary and display information for each role.**

➤ Input

✧ Employee.java

```
package com.mycompany.employeeproject;

public abstract class Employee {

    abstract void displayInfo();

    abstract void calculateSalary();

}
```

✧ Manager.java

```
package com.mycompany.employeeproject;

public class Manager extends Employee {

    void displayInfo() {

        System.out.println("--- Manager Info ---");

        System.out.println("Name : Shan");

        System.out.println("ID : 72387");

    }

}
```

```

    }

    void calculateSalary() {

        double baseSalary = 150000;

        double bonus = 30000;

        double totalSalary = baseSalary + bonus;

        System.out.println("Salary : " + totalSalary);

    }

}

```

✧ **Programmer.java**

```

package com.mycompany.employeeproject;

public class Programmer extends Employee {

    void displayInfo() {

        System.out.println("--- Programmer Info ---");

        System.out.println("Name : Saman&Shahri");

        System.out.println("ID : 54326");

    }

    void calculateSalary() {

        double hourlyRate = 500;

        double hoursWorked = 200;

        double totalSalary = hourlyRate * hoursWorked;

        System.out.println("Salary : " + totalSalary);

    }

}

```

✧ **EmployeeProject.java**

```

package com.mycompany.employeeproject;

public class EmployeeProject {

    public static void main(String[] args) {

        Employee m = new Manager();

        m.displayInfo();

        m.calculateSalary();

    }

}

```

```

        System.out.println();

        Employee p = new Programmer();

        p.displayInfo();

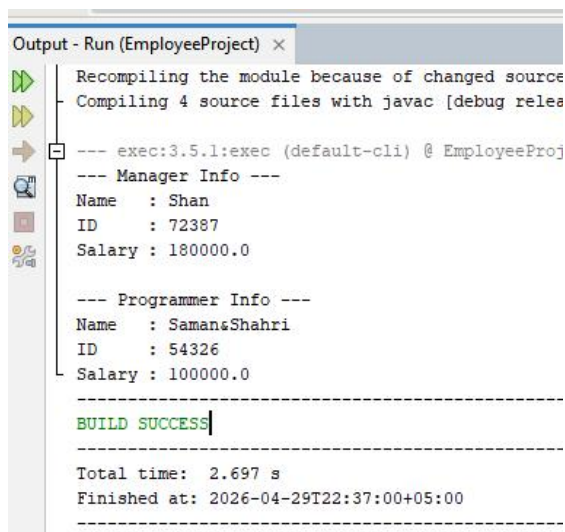
        p.calculateSalary();

    }

}

```

➤ Output



```

Output - Run (EmployeeProject) x
Recompiling the module because of changed source
Compiling 4 source files with javac [debug relea
--- exec:3.5.1:exec (default-cli) @ EmployeeProj
--- Manager Info ---
Name   : Shan
ID      : 72387
Salary : 180000.0

--- Programmer Info ---
Name    : Saman&Shahri
ID       : 54326
Salary  : 100000.0

-----
BUILD SUCCESS
-----
Total time:  2.697 s
Finished at: 2026-04-29T22:37:00+05:00
-----

```

- ✧ **Q3:** Write a Java program to create an abstract class **BankAccount** with abstract methods **deposit()** and **withdraw()**. Create subclasses: **SavingsAccount** and **CurrentAccount** that extend the **BankAccount** class and implement the respective methods to handle deposits and withdrawals for each account type.

➤ Input

✧ **BankAccount.java**

```

package com.mycompany.bankproject;

public abstract class BankAccount {

    abstract void deposit(double amount);

    abstract void withdraw(double amount);

}

```

✧ SavingsAccount.java

```
package com.mycompany.bankproject;

public class SavingsAccount extends BankAccount {

    String accountHolder = "Shan";

    int accountNumber = 72387;

    double balance = 50000;

    double minimumBalance = 10000;

    void displayInfo() {

        System.out.println("===== Savings Account =====");

        System.out.println("Name          : " + accountHolder);

        System.out.println("Account No    : " + accountNumber);

        System.out.println("Minimum Balance : " + minimumBalance);

        System.out.println("Current Balance : " + balance);

    }

    void deposit(double amount) {

        balance = balance + amount;

        System.out.println("Savings Deposit : +" + amount);

        System.out.println("Current Balance : " + balance);

    }

    void withdraw(double amount) {

        if (balance - amount < minimumBalance) {

            System.out.println("Withdrawal FAILED! Minimum balance " + minimumBalance + " rehna zaroori hai.");

        } else {

            balance = balance - amount;

            System.out.println("Savings Withdrawal : -" + amount);

            System.out.println("Current Balance : " + balance);

        }

    }

}
```

```

    }
}
}

```

✧ **CurrentAccount.java**

```

package com.mycompany.bankproject;

public class CurrentAccount extends BankAccount {

    String accountHolder = "Global Link Traders";

    int accountNumber = 10001;

    double balance = 20000;

    double overdraftLimit = 15000;

    void displayInfo() {

        System.out.println("===== Current Account =====");

        System.out.println("Name          : " + accountHolder);

        System.out.println("Account No    : " + accountNumber);

        System.out.println("Overdraft Limit : " + overdraftLimit);

        System.out.println("Current Balance : " + balance);

    }

    void deposit(double amount) {

        balance = balance + amount;

        System.out.println("Current Deposit : +" + amount);

        System.out.println("Current Balance : " + balance);

    }

    void withdraw(double amount) {

        if (balance - amount < -overdraftLimit) {

            System.out.println("Withdrawal FAILED! Overdraft limit " + overdraftLimit + " cross ho
            jayegi.");

        } else {

            balance = balance - amount;

            System.out.println("Current Withdrawal : -" + amount);

        }

    }

}

```

```
        System.out.println("Current Balance : " + balance);
    }
}
}
```

✧ **BankProject.java**

```
package com.mycompany.bankproject;

public class BankProject {

    public static void main(String[] args) {

        SavingsAccount sa = new SavingsAccount();

        sa.displayInfo();

        System.out.println();

        sa.deposit(5000);

        sa.withdraw(10000);

        sa.withdraw(40000);


        System.out.println();

        CurrentAccount ca = new CurrentAccount();

        ca.displayInfo();

        System.out.println();

        ca.deposit(8000);

        ca.withdraw(30000);

        ca.withdraw(20000);

    }

}
```

➤ Output

```
Output - Run (BankProject) x
--- resources:3.5.1:resources (default-resources) @ BankProject ---
skip non existing resourceDirectory C:\Users\hpzbo\OneDrive\Documents'

--- compiler:3.13.0:compile (default-compile) @ BankProject ---
Recompiling the module because of changed source code.
Compiling 4 source files with javac [debug release 25] to target\clas:

--- exec:3.5.1:exec (default-cli) @ BankProject ---

===== Savings Account =====
Name           : Shan
Account No     : 72387
Minimum Balance : 10000.0
Current Balance : 50000.0

Savings Deposit : +5000.0
Current Balance : 55000.0
Savings Withdrawal : -10000.0
Current Balance : 45000.0
Withdrawal FAILED! Minimum balance 10000.0 rehna zaroori hai.

===== Current Account =====
Name           : Global Link Traders
Account No     : 10001
Overdraft Limit : 15000.0
Current Balance : 20000.0

Current Deposit : +8000.0
Current Balance : 28000.0
Current Withdrawal : -30000.0
Current Balance : -2000.0
Withdrawal FAILED! Overdraft limit 15000.0 cross ho jayegi.

-----
BUILD SUCCESS
-----

Total time: 2.571 s
```